

DETAILED DESCRIPTION OF THE INVENTION

Comparative Modeling, Standardization, and Population Referencing

In various embodiments, the generated user-specific penile anatomical geometry profile may be further processed through comparative modeling systems configured to contextualize measured parameters relative to stored reference datasets. Such reference datasets may comprise anonymized geometry profiles, statistical distributions of structural parameters, or standardized anatomical reference models derived from aggregated measurements.

The system may compute normalized indices by comparing one or more measured structural metrics against corresponding reference distributions. Normalization may include percentile ranking, z-score transformation, ratio comparison, dimensional clustering, or other statistical mapping techniques. Derived indices may represent proportionality classifications, taper classifications, curvature classifications, volumetric bands, or other standardized structural descriptors.

In certain embodiments, the system may implement clustering algorithms configured to group geometry profiles into defined structural categories based on multidimensional similarity metrics. Similarity scoring may be calculated using Euclidean distance metrics, weighted parameter vectors, parametric matching functions, or machine learning classification models trained on stored geometry datasets.

The comparative modeling engine may further enable temporal comparison of a user's geometry profile across multiple measurement sessions. Longitudinal analysis may detect structural changes, stability trends, or measurement consistency over time. The system may generate delta metrics representing deviations from prior measurements, reference baselines, or population averages.

In some implementations, standardized geometry representations may be formatted into abstracted structural signatures suitable for integration with external systems, including compatibility modeling frameworks, device calibration engines, personalization algorithms, or anonymized research databases. Such standardized representations may exclude explicit raw geometry images while preserving parametric structural characteristics.

The comparative modeling and standardization architectures described herein are not limited to any specific statistical method, clustering approach, normalization technique, or reference dataset structure, provided that the system remains capable of contextualizing and interpreting a reconstructed penile anatomical geometry profile within a defined comparative framework.